LEGITMATE

A Neuro-Symbolic Legal Intelligence Framework

Property and civil document understanding, Evidence-Claim reasoning, temporal knowledge graphs, and Hybrid GraphRAG

PROJECT STATUS AND TECHNICAL REVIEW

Local research prototype | 31 October 2026

Research prototype only. Not legal advice or a validated legal decision system. Extracted statements, dates, entities, conflicts, and generated summaries require review against the original source.

Executive summary

LEGITMATE is a local-first prototype for organizing property and civil documents and tracing candidate findings back to source passages. It combines transparent extraction rules and human review states with pretrained multilingual retrieval models. The workflow supports document ingestion, hybrid search, cited answers, evidence review, timelines, a read-only knowledge graph, case briefs, and export bundles.

The strongest defensible result is traceability: findings can be linked to a file and source page or section, then reviewed and recorded. A source match confirms that text is present in an indexed source; it does not prove the underlying statement is true. The application exposes human review and abstention limitations rather than presenting output as a legal conclusion.

The current synthetic retrieval run covered 50 questions over three synthetic PDFs. All 30 answerable questions retrieved a labeled passage in the top five. Unsupported-query abstention was weak: BM25 abstained on 4 of 20 unsupported questions, while semantic, hybrid, graph, and reranked modes abstained on none. This is not evidence of real case performance.

Project objectives

- Keep passages and extracted candidates attached to a source location.
- Combine lexical search, multilingual semantic retrieval, rank fusion, optional reranking, and bounded graph expansion.
- Give reviewers explicit controls to assess support, contradiction, ambiguity, and unresolved evidence.
- Expose dates, intervals, candidate conflicts, and graph links as review aids with source context.
- Keep file processing local by default and disclose when optional external AI sends text to a provider.

System architecture

The browser calls a local FastAPI service. Readers extract page, slide, sheet, or section text. The system creates traceable passages and local indexes for lexical and semantic retrieval. Candidate extraction and evidence structures are distinct from answer drafting, so a retrieved or repeated statement is not automatically treated as true.

Layer	Implementation	Output and limit
Ingestion	PDF, DOCX/DOCM, PPTX, XLSX, TXT, Markdown, CSV, JSON, HTML, RTF	Source-located text; up to 200 MB per file
Retrieval	BM25, multilingual E5, reciprocal-rank fusion, optional cross-encoder	Ranked passages; ranking is not legal confidence
Extraction	Rules for entities, property identifiers, dates, amounts, and claims	Candidate items with source context; human review required
Evidence graph	Exact normalized entity mention links and bounded graph expansion	Navigation aid; no alias resolution or proof of relationship
Review and export	Evidence judgments, timeline, brief, CSV, and ZIP case bundle	Human annotations and source manifest; not legal opinion

Document workflow

1. Sign in to a local account workspace and optionally choose a matter.
2. Upload a supported file. Native text is read directly; PDF OCR is opt-in and local.
3. Review language estimate, coverage report, candidate findings, and citations.
4. Search with BM25, semantic, hybrid, graph-assisted, or reranked retrieval; open cited source passages.
5. Review claims and candidate facts, attach related evidence, and record a judgment or note.
6. Inspect timeline and graph context, then export reports or a case bundle.

Neuro-symbolic retrieval

The neural retrieval component uses the pretrained intfloat/multilingual-e5-small encoder when available. A general cross-encoder may rerank hybrid candidates. BM25 remains a lexical baseline, and a TF-IDF/LSA fallback can operate when E5 is unavailable. Reciprocal-rank fusion combines rankings; graph expansion adds bounded passages when extracted entity mentions match exactly.

The symbolic components include pattern-based candidate extraction, source text matching, reviewer states, typed date cues, and explicit interval links. Potential conflicts are deterministic review signals. No target-domain fine-tuned legal model is deployed. The GRU teaching demo is separate and is not used to answer document questions.

Evidence-Claim review

Claims can be associated with source passages and reviewer-selected evidence items. Reviewers can record supports, contradicts, ambiguous, or no direct support. These are human annotations, not automated legal conclusions. New uploads receive a SHA-256 fingerprint; a hash-chained local audit ledger records review action metadata.

Temporal context

Dates are extracted as candidates with rule-based event cues. Explicit from-to and between-and phrases can form interval links. Timeline ordering and graph date links help navigation but do not establish legal effect, causation, ownership, or a valid claim.

Automatic language detection and local OCR

The upload form no longer asks the user to choose an OCR language. Text-bearing files receive a local language estimate. When OCR is requested for a scan, LEGITMATE maps a sufficiently clear estimate to an installed Tesseract pack; scan-only PDFs can use Tesseract script detection when OSD data is available. If a detected pack is missing, the report records the local fallback.

Script identification is not full language identification. Latin-script scans fall back to English unless readable text provides a stronger estimate. OCR requires Tesseract and local language data. Tesseract was not available on PATH in the checked environment, so scanned-only OCR could not be exercised here. New regression cases cover Hindi text detection, missing-pack fallback, and scan-only script detection.

Security and privacy posture

The project uses signed sessions, CSRF protection, account-local workspaces, matter filters, source hashes, and a hash-chained metadata audit log. It has not completed independent security review, threat modeling, deployment review, backup/restore testing, or a privacy impact assessment. Optional Gemini or Groq features send the request or matching excerpts described in the app privacy notice; external AI is off unless configured and selected.

Current export formats

The web app produces per-document NLP reports as HTML, an evidence matrix as CSV, and a case bundle as ZIP. The accompanying PowerPoint deck is a project presentation artifact; it does not mean the web app exports project slides.

Recorded retrieval evaluation

The current benchmark uses 50 questions (30 answerable, 20 unsupported) over three bundled synthetic PDFs and six pages. In the recorded run, every mode retrieved a labeled passage in the top five for every answerable question.

Mode	MRR@5	nDCG@5	Hit@5	Unsupported abstention
BM25	0.917	0.928	1.000	0.200 (4/20)
Semantic	0.861	0.895	1.000	0.000 (0/20)
Hybrid	0.900	0.924	1.000	0.000 (0/20)
Graph	0.917	0.936	1.000	0.000 (0/20)
Reranked	0.933	0.949	1.000	0.000 (0/20)

The unsupported-query results show a known weakness. In this run, semantic, hybrid, graph, and reranked modes answered every unsupported query. Do not tune production thresholds on this small synthetic set or interpret supported-citation measures as real-world citation accuracy.

Recorded extraction evaluation

The bundled extraction benchmark was run on the current code: three synthetic PDFs and six pages. It matched the small labeled set with 1.000 precision and recall for people, organizations, property identifiers, dates, amounts, and claims, and found both known conflict cases. The small sample is not a real-world accuracy estimate. The separate GRU teaching model scored 0.750 on 12 held-out synthetic examples and is not connected to document answers.

Regression verification

Eight local unit tests pass in this build: four extraction-quality checks and four automatic OCR-language selection checks. Test cases cover the chosen rules and fallbacks; they do not validate legal correctness or OCR transcription quality.

Limitations and research plan

No independently reviewed real-world property/civil corpus is bundled. Retrieval, extraction, OCR, multilingual performance, citation accuracy, and security posture need further evaluation before broader use.

Recommended next steps

1. Collect authorized, protected property and civil documents with documented provenance and license or consent.

2. Have two reviewers independently label spans, adjudicate disagreements, and split by matter to prevent leakage.

3. Measure entity/date/amount/claim precision, recall, and F1 by document, language, script, and OCR status.

4. Measure retrieval hit/recall, MRR/nDCG, citation-span accuracy, and unsupported-query abstention on held-out matters.

5. Audit access controls, retention, encryption, backups, dependency risks, deployment settings, and incident handling.

Reproduction and demonstration

On Windows, install project requirements and configure a unique administrator password and random session secret in the private .env file. Start the app from the repository root with .\run_legitmate.ps1, keep the terminal open, and visit http://127.0.0.1:8000. Use the bundled synthetic PDFs in data/sample for a reproducible demo. Show citations, reviewer judgments, timeline, graph, and an unsupported query.

Project references

- LEGITMATE README.md: product scope, modules, limitations, and evaluation results.
- data/evaluation/REVIEWED_CORPUS_GUIDE.md: annotation and held-out evaluation protocol.
- docs/REVIEWER_WALKTHROUGH.md: reproducible demo and interpretation guidance.
- tests/test_extraction_quality.py and tests/test_ocr_language.py: local regression checks.

Conclusion

LEGITMATE is a substantial local research prototype with source-linked candidate extraction, hybrid retrieval, human review controls, graph and timeline views, and export workflows. Its defensible role is helping users organize and inspect evidence with citations. A credible legal-accuracy claim requires independent real-document evaluation, stronger unsupported-answer handling, OCR validation, and security review.